

ASSIGNMENT 1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend Development

Frontend is the part of a website that users can see and use. It includes buttons, images, menus, and text. A frontend developer makes the website look attractive and easy to use.

Example: On Amazon, the search bar, product images, and "Add to Cart" button are part of the frontend.

Backend Development

Backend is the part that works behind the scenes. It stores data, manages users, and processes requests. Users cannot see this part.

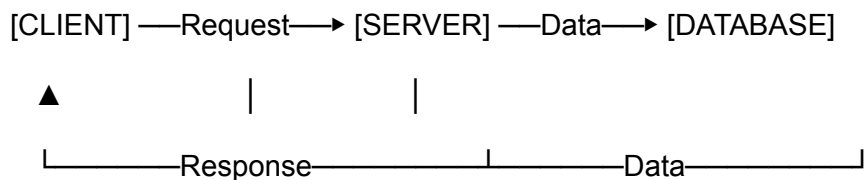
Example: When you log in to Amazon or place an order, the backend checks your details and saves your order information.

Full-Stack Development

A full-stack developer works on both the frontend and the backend. They can build a complete website from start to finish.

Example: A full-stack developer can create an online shopping website by designing the pages as well as handling the login, database, and payment system.

2. Create a simple diagram showing how the client-server model works in web architecture.



Client: Sends request.

Server: Processes request.

Database: Stores and provides data.

Response: Server sends data back to the client.

3. Describe how a browser requests and displays a web page from a web server.

1. The user enters a website URL or clicks a link in the web browser.
2. The browser sends an HTTP/HTTPS request to the web server asking for the web page.
3. The web server processes the request, retrieves the required files or data, and sends them back to the browser.
4. The browser receives the HTML, CSS, and JavaScript files, processes them, and displays the web page on the screen so the user can interact with it.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

1. **VS Code (Code Editor):**
Used to write and edit HTML, CSS, and JavaScript code easily.
2. **Web Browser (Google Chrome/Edge):**
Used to open, test, and check how the website looks and works.
3. **Git:**
Helps track changes in your project and saves different versions of your code.
4. **GitHub:**
Used to store projects online, share code, and work with others.
5. **Node.js:**
Allows JavaScript to run outside the browser and supports many web development tools.
6. **npm (Node Package Manager):**
Used to install and manage packages, libraries, and frameworks.
7. **Live Server (VS Code Extension):**
Automatically refreshes the webpage whenever you save changes, making development faster.

Conclusion:

These tools help developers write, test, organize, and manage web development projects easily and efficiently.

5. Explain what a web server is and give examples of commonly used servers.

A web server is a software or computer that stores website files and sends them to users when they open a website in a browser.

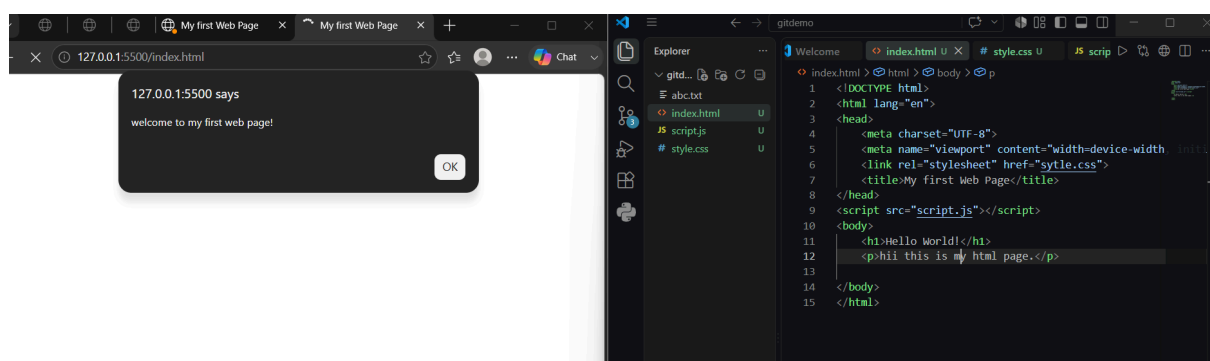
Common examples of web servers:

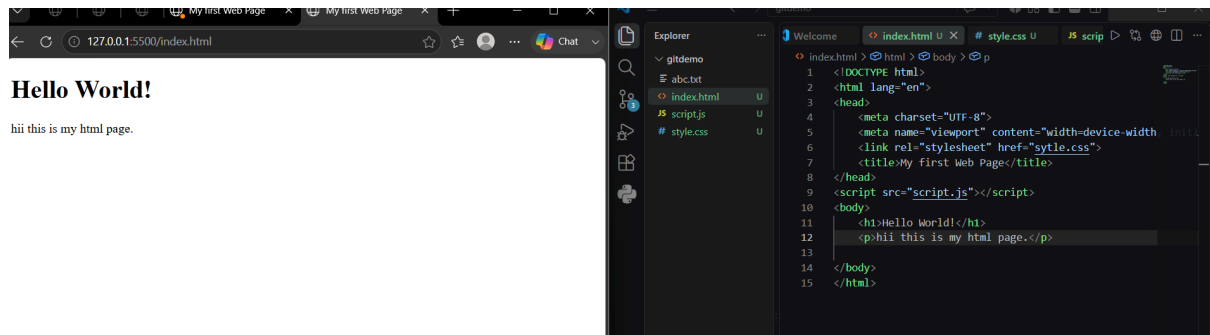
- Apache – One of the most popular and widely used web servers.
- Nginx – Known for its speed and handling many users at once.
- Microsoft IIS – A web server developed by Microsoft for Windows.
- LiteSpeed – A fast and secure web server often used for high-performance websites.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

- Frontend Developer:
Creates the part of the website that users see and interact with, such as buttons, menus, and layouts.
- Backend Developer:
Builds the server-side logic, manages data, and makes sure the website works properly behind the scenes.
- Database Administrator (DBA):
Manages the database by storing, organizing, securing, and maintaining the project's data efficiently.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.





8. Explain the difference between static and dynamic websites. Provide an example of each.

- Static Website:

A static website shows the same content to every user. The content only changes when the developer edits the website files.

Example: A personal portfolio website or a college project website.

- Dynamic Website:

A dynamic website shows different content based on the user or data stored in a database. The content updates automatically without changing the website files.

Example: Instagram, Amazon, or Facebook.

- Difference:

A static website has fixed content, while a dynamic website displays updated or personalized content based on user interaction or database information.

9. Research and list five web browsers. Explain how rendering engines differ between them.

Five Web Browsers:

1. Google Chrome – Uses the Blink rendering engine.
2. Microsoft Edge – Uses the Blink rendering engine.
3. Mozilla Firefox – Uses the Gecko rendering engine.
4. Safari – Uses the WebKit rendering engine.
5. Opera – Uses the Blink rendering engine.

- A rendering engine is responsible for displaying web pages in a browser.
- Different browsers use different rendering engines, such as Blink, Gecko, and WebKit.
- Because of this, a website may look or perform slightly differently in different browsers.
- Each rendering engine has its own strengths in terms of speed, compatibility, and performance.
- Developers test websites in multiple browsers to ensure they work correctly for everyone.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

